FPT UNIVERSITY
DATA SCIENCE METHODOLOGY

10-Step SQL & Python Analytics Pipeline

6,607 Student Records
16 Attributes
13 SQL Queries & EDA

Final Project Defense
Student Performance Factors Analysis

Subject: Data Science with Python & SQL (ADY201m)
Dataset: StudentPerformanceFactors.csv (n = 6,607)

Key Highlights:
- Complete 10-Step Data Science Methodology (SQL + Python)
- Non-destructive SQL Data Cleaning (CTE & Window Functions)
- Exploratory Data Analysis (EDA) & IQR Outlier Detection
- Statistical Testing (ANOVA / T-test) & Linear Regression

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AGENDA

Presentation Outline

01. Business Understanding & Approach

• Steps 1–2: Problem formulation, analytical questions, SQL + Python methodology.

02. Data Requirements & Collection

• Steps 3–4: 6,607 records, 16 columns, rowid key generation, initial SQL queries.

03. Data Quality & SQL/Python Cleaning

• Steps 5–6: Handling missing values (Unknown) and out-of-range score (101 -> NULL).

04. Exploratory Data Analysis (EDA)

• Distribution of Exam_Score, scatter fits, and Pearson correlation matrix.

05. Categorical Factors & Demographics

• Violin/Box plots across family, school, peer, gender, and school type.

06. SQL Rule-Based Modeling (Step 7)

• CASE WHEN study tiers, RANK() OVER window function, Motivation × Parent matrix.

07. ANOVA Testing & Linear Regression

• Hypothesis testing (8/10 significant factors) and multivariate regression (R^2 = 85.8%).

08. Evaluation, Deployment & Feedback

• Steps 8–10: SQL JOIN cohort evaluation, CSV/DB export, and policy recommendations.

01

STEPS 1 & 2 — METHODOLOGY

Business Understanding & Analytic Approach

Step 1: Business Understanding (Problem)

- Educational institutions need to identify which study behaviors and socioeconomic contexts accompany high or low Exam_Score.
- Goal: Prioritize academic advising and resource support for at-risk student groups.
- Focus on associative patterns and actionable cohort benchmarks rather than causal claims.

Core Analytical Questions

- Q1: What is the average exam score (67.23) and overall data quality profile?
- Q2: How do scores differ across weekly study-hour tiers (<=10h to >30h) and attendance brackets (<70% to 90–100%)?
- Q3: How do motivation, parental involvement, parental education, and extracurriculars interact?

Step 2: SQL Analytic Approach

- Data quality audit: COUNT, GROUP BY HAVING, NULL/empty checks, and range validation.
- Non-destructive cleaning via CTE (WITH standardized AS) & ROW_NUMBER() deduplication.
- Rule-based cohort modeling using CASE WHEN, RANK() OVER, SUM() OVER, and CROSS/INNER JOIN.

Step 2: Python Statistical & ML Approach

- Pandas & NumPy pipeline to cross-verify 100% of SQL cleaning and group aggregations.
- Exploratory Data Analysis (EDA) with Matplotlib/Seaborn and IQR outlier detection.
- Parametric hypothesis testing (One-way ANOVA, T-test) and Multivariate Linear Regression.

Combining SQL relational querying with Python statistical modeling ensures 100% reproducibility.

02

STEPS 3 & 4 — DATA PIPELINE

Data Requirements & SQL Data Collection

Step 3: Dataset Architecture (6,607 × 16)

- Source: StudentPerformanceFactors.csv (6,607 rows, 16 columns).
- Primary Key Generation: Raw CSV has no ID column -> generated student_id = rowid (1 to 6,607).
- Target Variable: Exam_Score (0–100 scale).
- 5 Numeric Predictors: Hours_Studied, Attendance, Previous_Scores, Sleep_Hours, Physical_Activity.
- 10 Categorical Predictors: Parental_Involvement, Access_to_Resources, Motivation_Level, Family_Income, Teacher_Quality, Peer_Influence, Parental_Education_Level, Extracurricular_Activities, School_Type, Gender.

student_id	Hours_Studied	Attendance	Exam_Score
1526	27	98%	101
3580	14	90%	99
4193	28	90%	98
6348	28	96%	98

Step 4: Initial SQL Retrieval Findings

- Query 4.1 (Top 10 students with Attendance >= 90%): Top 4 rows above immediately uncover student_id = 1526 with Exam_Score = 101.
- Basis for Steps 5–6 cleaning: Flagged the out-of-range score (101 -> NULL) without altering or dropping raw records.
- Verified with Pandas: 100% match across the Top 10 high-attendance cohort and student_id order.

SQLite table raw_students preserves the exact original CSV for auditability before cleaning.

03

STEPS 5 & 6 — DATA CLEANING

Data Quality Audit & SQL/Python Cleaning

Quality Metric	Before (Raw)	After (Clean SQL/Py)
Total Records	6,607	6,607 (student_id 1..6607)
Duplicate Rows	0	0 (ROW_NUMBER rn = 1)
Missing Teacher_Quality	78 (1.18%)	0 (78 -> 'Unknown')
Missing Parental_Education	90 (1.36%)	0 (90 -> 'Unknown')
Exam_Score > 100 (Invalid)	1 (id=1526, score=101)	0 (1 -> NULL, n=6,606 valid)
Valid Mean Exam_Score	67.236	67.231

5 Golden Rules of Step 6 Cleaning

- 1. Generate primary key student_id = rowid.
- 2. Standardize column names to snake_case and TRIM() text.
- 3. Never fabricate exam scores: out-of-range Exam_Score = 101 is flagged as NULL rather than capped to 100.
- 4. Map missing categorical entries to 'Unknown' via COALESCE(NULLIF(TRIM(col), ''), 'Unknown').
- 5. Execute cleaning in new table students_clean_sql & DataFrame clean_py without altering raw_students.

Reconciliation Verification (SQLite vs. Pandas)

- All automated assertions passed: len(clean_sql) == len(clean_py) == 6,607 | mean(exam_score) == 67.231 | Unknown counts (78 & 90) match 100%.

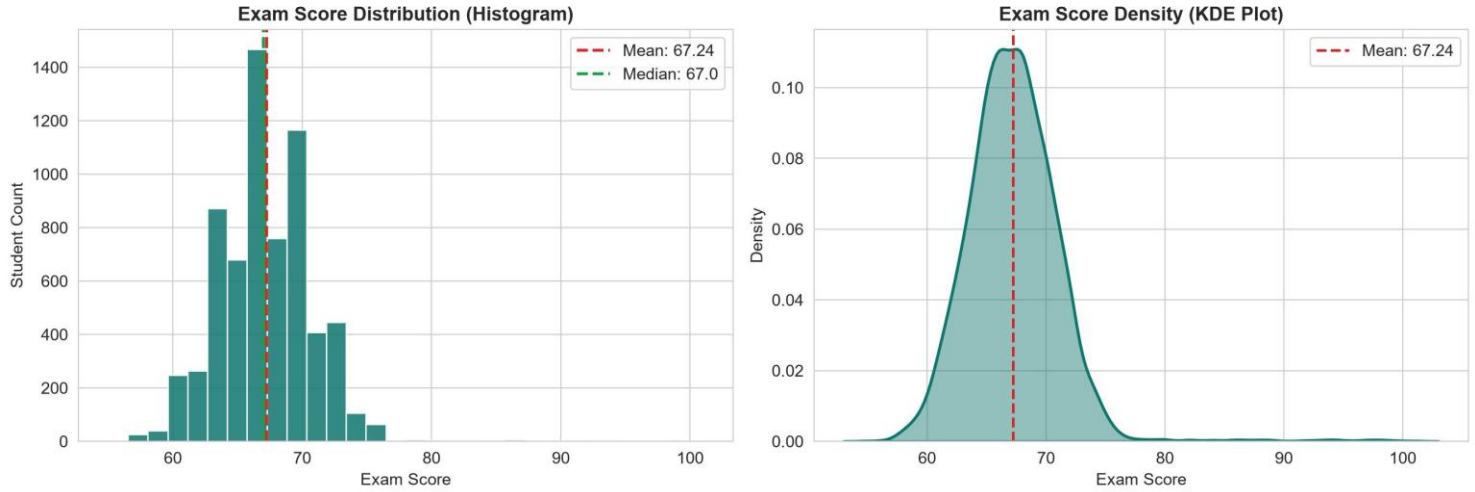
Non-destructive data preparation ensures full transparency and prevents artificial bias.

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EXPLORATORY DATA ANALYSIS

Exam Score Distribution & Outlier Audit

Figure 1: Target Exam Score Distribution (n = 6,607)



Key Distributional Insights

- Mean = 67.24 | Median = 67.0 | Std = 3.89.
- IQR = 4.0 (Q1 = 65.0, Q3 = 69.0): 50% of students score between 65 and 69.
- Right-skewed distribution (Skewness = 1.645, Kurtosis = 10.575) driven by high achievers scoring 76–100.
- IQR Outliers: Hours_Studied has 43 outliers (0.65%), Exam_Score has 104 high-achieving outliers (1.57%); Attendance, Sleep_Hours, Previous_Scores, and Physical_Activity have 0% outliers.

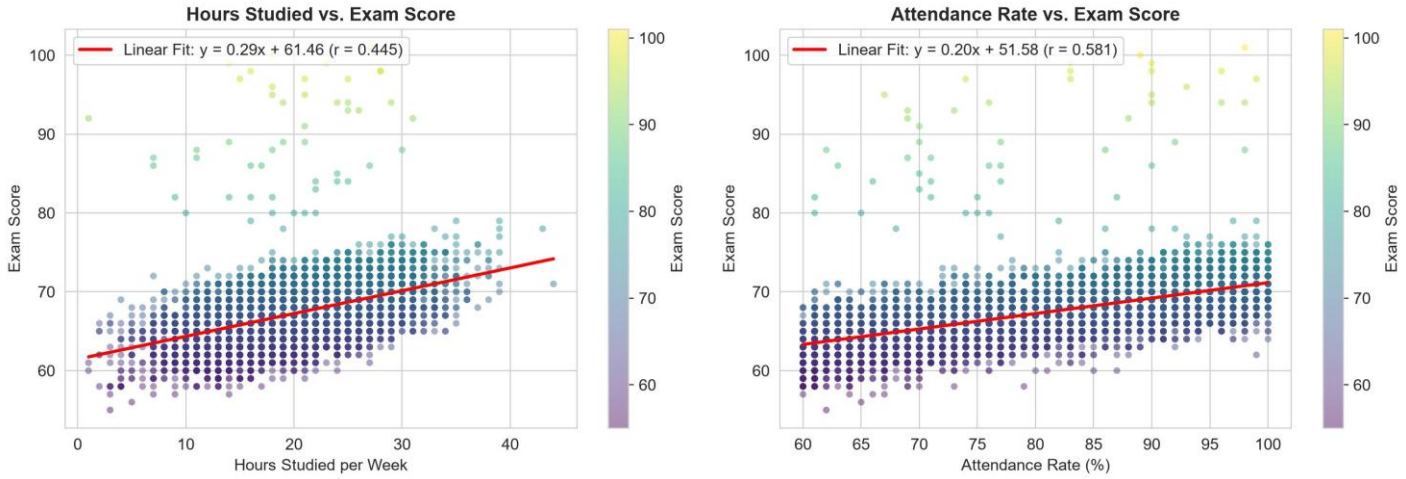
Predictor variables are symmetrically distributed, while Exam_Score exhibits a high-achieving right tail.

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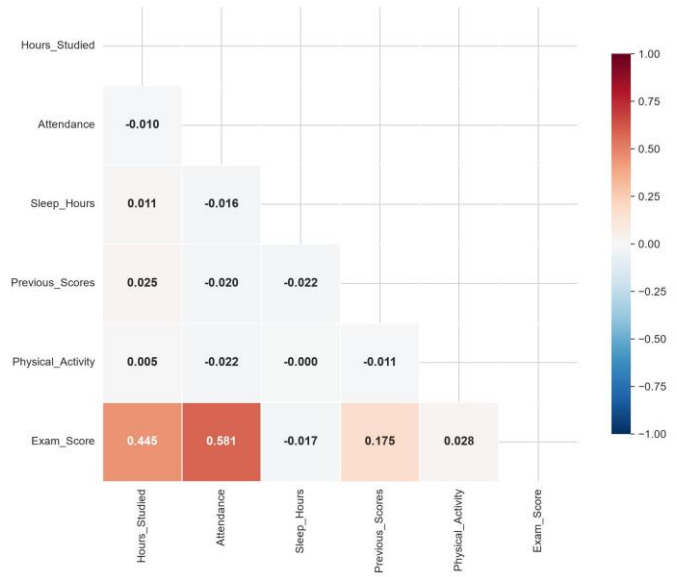
EXPLORATORY DATA ANALYSIS

Key Numerical Drivers & Correlation Matrix

Figure 2: Relationship Between Study Hours, Attendance, and Exam Score



HÌNH 3: Ma trận tương quan giữa các biến số



Pearson Correlation Summary (r with Exam_Score)

- 1. Attendance (r = +0.581, strong positive): Each +10% attendance corresponds to +1.98 exam score points (y = 0.20x + 51.59).
- 2. Hours_Studied (r = +0.445, moderate-strong positive): Each +10 study hours/week adds +2.90 exam score points (y = 0.29x + 61.46).
- 3. Previous_Scores (r = +0.175, weak positive) | Physical_Activity (r = +0.028, negligible) | Sleep_Hours (r = -0.017, negligible).

Attendance and weekly study hours are the two dominant numerical drivers of student exam performance.

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EXPLORATORY DATA ANALYSIS

Impact of Categorical & Demographic Factors

Figure 4: Impact of Key Categorical Factors on Exam Score (ANOVA p < 0.001)

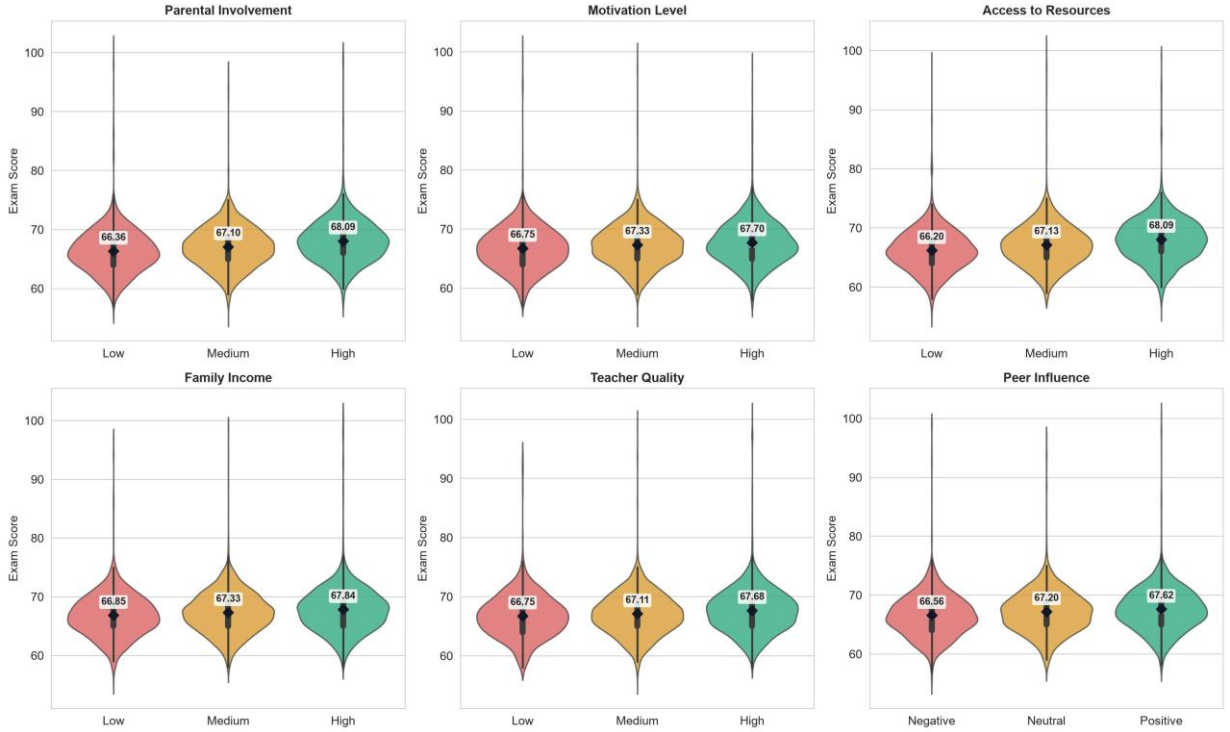
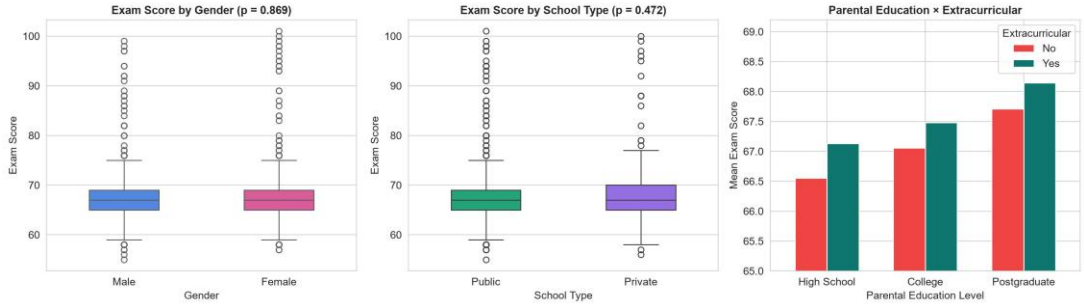


Figure 5: Score Comparison Across Demographics and Extracurricular Participation



Categorical Group Comparisons

- Access_to_Resources: High (68.09) vs. Low (66.20) -> Diff = +1.89 pts.
- Parental_Involvement: High (68.09) vs. Low (66.36) -> Diff = +1.73 pts.
- Parental_Education: Postgraduate (67.97) vs. High School (66.89) -> +1.08 pts.
- Peer_Influence: Positive (67.62) vs. Negative (66.56) -> +1.06 pts.
- Gender (67.24 vs 67.23) & School_Type (67.29 vs 67.21): No meaningful difference.

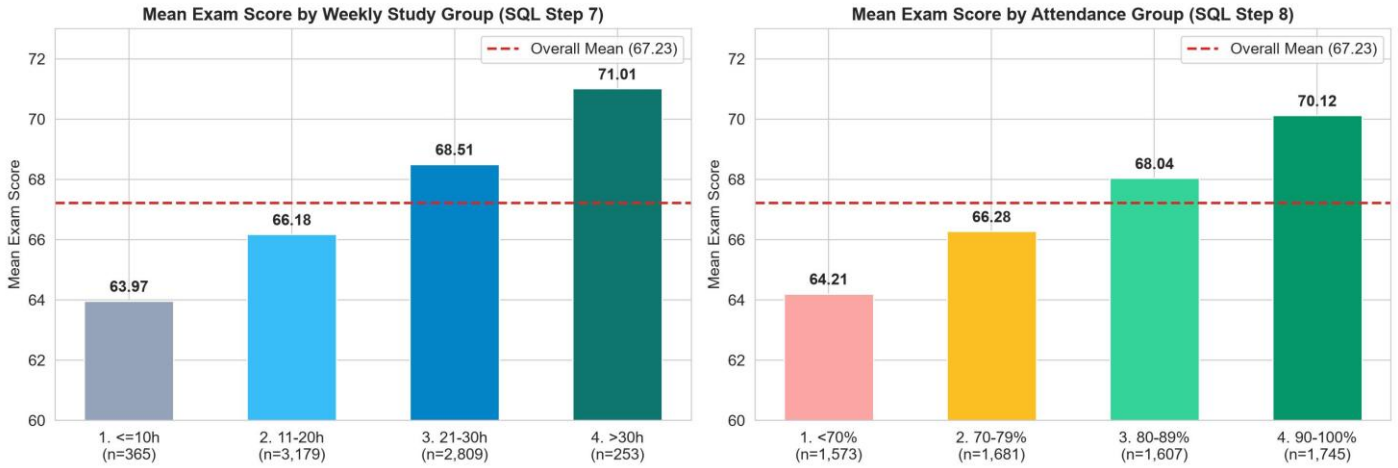
Supportive home environments, school resources, and positive peer networks consistently elevate student scores.

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STEPS 7 & 8 — SQL MODELING

SQL Rule-Based Segmentation (Study & Attendance)

Figure 6: SQL Rule-Based Segmentation by Study Hours and Attendance Rate



Study Group	Count	Avg Score	Diff	Rank
1. <=10h	365	63.97	-3.26	4
2. 11-20h	3,179	66.18	-1.05	3
3. 21-30h	2,809	68.51	+1.28	2
4. >30h	253	71.01	+3.78	1

Attendance Tier	Students	Avg Score	vs. <70%
1. < 70%	1,573	64.21	Baseline
2. 70 - 79%	1,681	66.28	+2.07
3. 80 - 89%	1,607	68.04	+3.83
4. 90 - 100%	1,745	70.12	+5.91

SQL Window Function & Segmentation Insights

- Students studying >30h/week score 71.01 on average (+7.04 pts higher than <=10h/week), even though average attendance (~79–80%) is nearly identical across study groups.
- Students with 90–100% attendance average 70.12 (+5.91 pts vs. <70%; n = 6,606 valid scores, excl. 1 NULL).

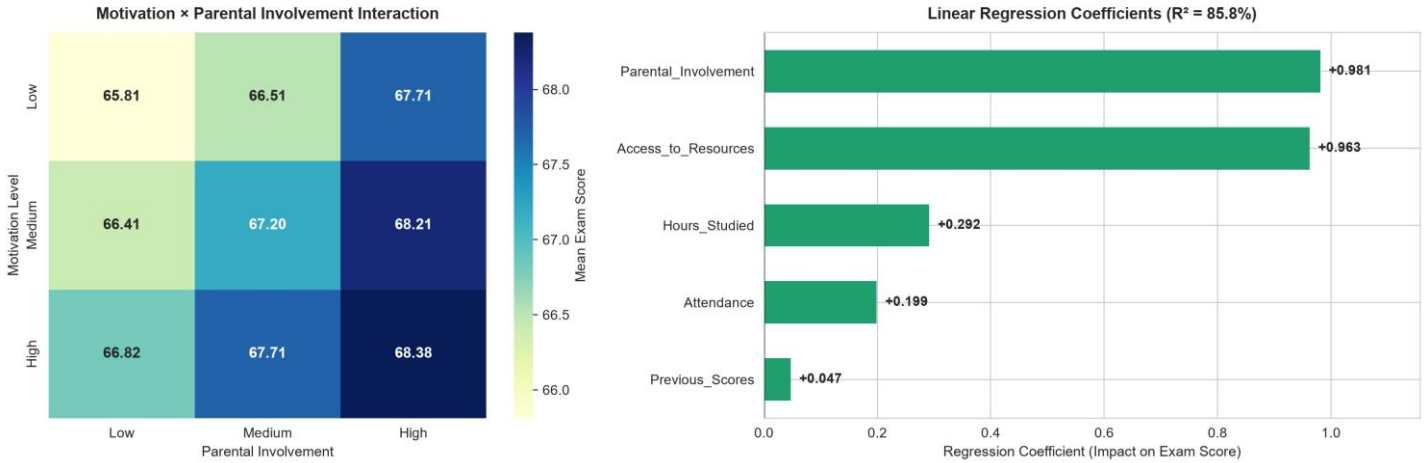
SQL CASE WHEN, CTEs, and RANK() OVER provide crystal-clear cohort benchmarks verified by Python.

08

STATISTICAL & ML MODELING

ANOVA Hypothesis Testing & Linear Regression

Figure 7: Multi-Factor Interaction Matrix and Regression Feature Importance



Interaction Effect & Regression Performance

- High Motivation × High Parental Involvement yields 68.38 vs. 65.81 for Low × Low (+2.57 pts).
- Linear Regression (5 key predictors, IQR [59, 75] outlier filter) achieves R^2 = 85.8% (up from 60.8% initially).

Variable	Test	Stat	p-value
Access_to_Resources	ANOVA	98.00	< 0.001
Parental_Involvement	ANOVA	84.49	< 0.001
Parental_Education	ANOVA	36.43	< 0.001
Peer_Influence	ANOVA	34.05	< 0.001
Family_Income	ANOVA	29.79	< 0.001
Motivation_Level	ANOVA	25.72	< 0.001
Teacher_Quality	ANOVA	19.64	< 0.001
Extracurricular	T-test	-5.24	< 0.001
School_Type	T-test	0.72	0.472 (NS)
Gender	T-test	0.17	0.869 (NS)

8 of 10 categorical factors are statistically significant (p < 0.001); Gender and School Type are not.

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STEPS 8, 9 & 10 — WRAP-UP

SQL JOIN Evaluation, Deployment & Conclusion

Parental Education	Extracurricular	Students	Avg Score	Diff
Postgraduate	Yes	774	68.15	+0.92
Postgraduate	No	531	67.71	+0.48
College	Yes	1,208	67.48	+0.25
High School	Yes	1,906	67.11	-0.12
College	No	781	67.06	-0.17
High School	No	1,317	66.55	-0.68

Step 9: Deployment Deliverables

- SQLite DB: student_analysis.db (raw_students, students_clean_sql, students_clean_python).
- Standalone SQL: TruyVan_LamSach_PhanTich.sql (13 queries).
- Cleaned CSVs: student_performance_cleaned_sql.csv, student_performance_cleaned_python.csv, analysis_study_groups.csv.

Step 10: Feedback & Strategic Recommendations

- 1. Data Governance: Keep non-destructive cleaning; verify anomalous record student_id = 1526 (Exam_Score = 101) with exam office.
- 2. Early Warning System: Prioritize academic intervention for students with Attendance < 75% or Hours_Studied <= 10h/week.
- 3. Family-School Partnership: Engage parents through regular progress reports, as High Parental Involvement + High Motivation adds +2.57 pts.
- 4. Resource Equity: Provide learning resources and extracurricular opportunities to low-income students to close the achievement gap.

Thank You for Listening! — Q&A Session (Data Science with Python & SQL - ADY201m)